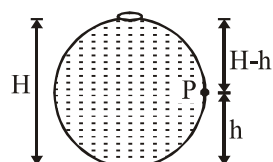
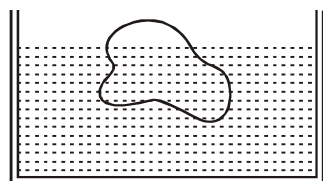


PROPERTIES OF MATTER & FLUID MECHANICS

- Equal masses of water and a liquid of relative density two are mixed together; then mixture has a relative density of :
(1) $2/3$ (2) $4/3$ (3) $3/2$ (4) 3
- A vessel contains a liquid of density ρ as shown in figure. The gauge pressure at the point P is :

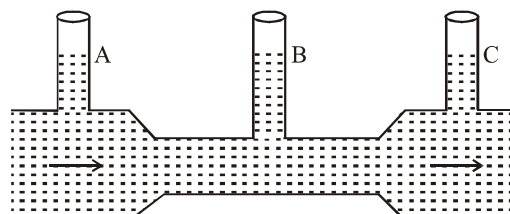


- $h\rho g$ (2) $H\rho g$
(3) $(H - h)\rho g$ (4) None of these
- When an air bubble comes from the bottom of a lake to the top surface, then its radius:
(1) Increases (2) Decreases
(3) Does not change (4) Becomes zero
- A body floats in a liquid contained in a beaker. The whole system as shown in figure falls freely under gravity. The upthrust on the body due to the liquid is :

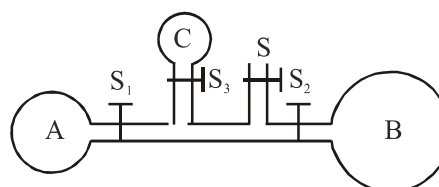


- zero
(2) Equal to the weight of the liquid displaced by the body
(3) Equal to the weight of the body in air
(4) Equal to the weight of the immersed portion of the body in the liquid
- A solid is completely immersed in an ideal liquid. The force exerted by the liquid on the solid will
(1) increase, if it is pushed deeper inside the liquid
(2) change, if its orientation is changed keeping it inside the liquid
(3) decrease, if it is taken partially out of the liquid
(4) All of these

- Three tubes A, B and C are connected to a horizontal pipe in which an ideal liquid is flowing. The radii of the tubes A, B and C at the junction are respectively 2 cm, 1 cm and 2 cm. It can be said :-

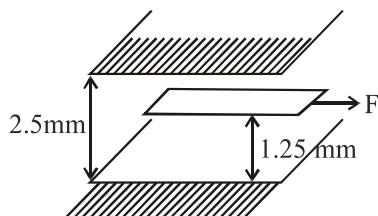


- The height of the liquid in the tube A is maximum
 - Height of the liquid in the tubes A and B is same
 - Height of the liquid in the tubes A, B and C is same
 - Height of the liquid in the A and C is same
- The given figure shows three soap bubbles A, B and C are prepared by blowing the capillary tube with stop cocks S , S_1 , S_2 and S_3 . With stop cock S closed and S_1 , S_2 and S_3 opened, then-



- C and A will both start collapsing with volume of B increasing
 - B will start collapsing with volumes of A and C increasing
 - C will start collapsing with volumes of A and B increasing
 - volumes of A, B and C will become equal at equilibrium
- A volume V of a viscous liquid flows per unit time due to pressure head of ΔP along a pipe of diameter D and length l . Instead of this pipe a set of four pipes each of diameter $D/2$ and length $2l$ is connected in parallel to the same pressure head of ΔP . Now the volume of the liquid flowing per unit time is -
(1) V (2) $V/4$ (3) $V/6$ (4) $V/8$

9. A space 2.5 mm wide between two large plane surfaces is filled with oil as shown in diagram. Force required to drag a very thin plate of area of one side 0.5 m^2 just midway of the surfaces at a velocity of 0.5 m/sec is 1 N . The coefficient of viscosity in kg/m-sec is -



- (1) 1×10^{-2} (2) 2.5×10^{-2}
 (3) 5×10^{-2} (4) 2.5×10^{-3}
10. Two spherical metal balls (of the same material) with radii 0.2 cm and 0.3 cm are falling through a fluid at constant speeds. If the speed of the first ball (radius = 0.2 cm) is 4 cm/s . What is the speed of the other ball ?
 (1) 6 cm/s (2) 8 cm/s (3) 9 cm/s (4) 12 cm/s
11. When water spreads on a glass plate, the force between the water molecules and glass molecules is called:-
 (1) Surface tension
 (2) Cohesive force
 (3) Adhesive force
 (4) None of these
12. Due to capillary action water rises to a height of 16.3 cm in a capillary of length 18 cm above the water level. If the tube is cut at height of 12 cm
 (1) Water will come as fountain from the capillary tube
 (2) Water will stay at a height of 12 cm in the capillary tube but radius of curvature at top will increase.
 (3) Water will stay at a height of 12 cm in capillary tube but radius of curvature at top will decrease.
 (4) Water will flow down the side of the capillary tube

13. Two identical air bubbles are rising in water. If they coalesce under isothermal condition to form a single air bubble, the combined air bubble would have -

- (1) Higher excess pressure but smaller terminal velocity
 (2) Higher excess pressure and higher terminal velocity
 (3) Lower excess pressure and smaller terminal velocity
 (4) Lower excess pressure but higher terminal velocity

14. Force constant (K) of a spring is analogous to -

- (1) $\frac{YA}{L}$ (2) $\frac{YL}{A}$ (3) $\frac{AL}{Y}$ (4) ALY

15. Which of the following statement(s) is/are correct?

- I. Atoms or molecules in a fluid are arranged in a random manner.
 II. A fluid can withstand tangential or shearing stress for an indefinite period.
 III. A fluid has no definite shape of its own.
 (1) Only I (2) Both I and III
 (3) Only III (4) All of these

16. Select the wrong definition :-

- (1) Deforming force – Force that changes configuration of body
 (2) Elasticity – Property of regaining original configuration
 (3) Plastic body – Which can be melted easily
 (4) Elastic limit – Beyond which material begins to flow

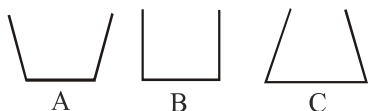
17. A change in pressure is applied at a particular point, in an enclosed fluid _____ to every point of the fluid and the walls of the containing vessel.

Fill in the blank -

- (1) Transmitted increasingly
 (2) Transmitted decreasingly
 (3) Transmitted undiminished
 (4) Transmitted randomly

18. The velocity of viscous fluid in contact with a surface is -
 (1) Same as that of the surface
 (2) More than that of the surface
 (3) Less than that of the surface
 (4) Always zero
19. Fluid does not have any -
 (1) Resistance to change in its volume
 (2) Resistance to change in its shape
 (3) Resistance to change in its pressure
 (4) None of the above
20. Pressure at a point inside a liquid at rests does not depend on -
 (1) The depth of the point below the surface of the liquid
 (2) The nature of the liquid
 (3) The acceleration due to gravity at that point
 (4) Total weight of the liquid in the beaker
21. We have three beakers A, B and C containing three different liquids. They are stirred vigorously and placed on a table. Then, liquid which is -
 (1) Most viscous comes to rest at the earliest
 (2) Most viscous comes to rest at the last
 (3) Most viscous slow down earliest but comes to rest at the last
 (4) All of them come to rest at the same time
22. As the temperature of water increases. Its viscosity.
 (1) Remain unchanges
 (2) Increases
 (3) Decreases
 (4) May increase or decrease
23. In a streamline flow -
 (1) The speed of a particle always remain same
 (2) The velocity of a particle always remain same
 (3) The velocities of particles arriving at a given point are same
 (4) The kinetic energy of a particle always remain same
24. The terminal velocity of an object in a viscous fluid is proportional to -
 (1) Sum of densities of both object and fluid
 (2) Difference of densities of both object and fluid
 (3) Density of object only
 (4) Density of viscous fluid only
25. For an aerofoil which of the following statement(s) is/are correct -
 (I) The orientation of the wings relative to flow direction causes the streamline
 (II) Flow speed on top is lower than that below it
 (III) Upward force resulting in dynamic lift of the wings
 (1) Only I
 (2) Only III
 (3) Only I and II
 (4) Only I and III
26. The viscosity of gases :
 (1) Increases with increase in temperature
 (2) Decreases with increase in temperature
 (3) Does not depend on temperature
 (4) Increase with increase in pressure
27. A body is moving through a viscous fluid then choose the incorrect option
 (1) Viscous force acting on the body depends on density of body
 (2) Viscous force acting on the body depends on viscosity of fluid
 (3) Viscous force acting on the body depends on velocity of body
 (4) Viscous force acting on the body is directed opposite to the direction of velocity of the body
28. The surface tension of a liquid at its boiling point -
 (1) Becomes zero
 (2) Becomes infinity
 (3) Is equal to the value of its surface tension at room temperature
 (4) Is half to the value of its surface tension at the room temperature

29. The three vessel shown in figure have same base area. Equal volumes of a liquid are poured in the three vessel. The force on the base will be-

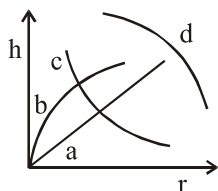


- (1) Maximum in vessel A
 (2) Maximum in vessel B
 (3) Maximum in vessel C
 (4) Equal in all the vessels
30. Equal masses of three liquids are kept in three identical cylindrical vessels A, B and C. The densities are ρ_A , ρ_B , ρ_C respectively where $\rho_A < \rho_B < \rho_C$ then the force on the base will be -
 (1) Maximum in vessel A
 (2) Maximum in vessel B
 (3) Maximum in vessel C
 (4) Equal in all the vessels
31. A Beaker containing a liquid is kept inside a big closed jar. If the air inside the jar is continuously pumped out, the pressure in the liquid near the bottom of the liquid will be :
 (1) Increase
 (2) Decrease
 (3) Remain constant
 (4) First decrease and then increase
32. The pressure in a liquid at two points in the same horizontal plane are equal. Consider an elevator accelerating upward and a car accelerating on a horizontal road then the above statement is correct for -
 (1) The car only
 (2) The elevator only
 (3) Both of them
 (4) Neither of them
33. A Barometer kept in an elevator accelerating upward reads 76 cm. The air pressure in the elevator is -
 (1) 76 cm
 (2) < 76 cm
 (3) > 76 cm
 (4) zero

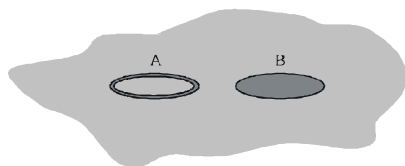
34. A piece of wood is floating in water kept in a bottle. The bottle is connected to an air pump. Neglect the compressibility of water. If more air is pushed into the bottle from the pump, the piece of wood will float with -

- (1) Larger part in the water
 (2) Lesser part in the water
 (3) Same part in the water as earlier
 (4) It will sink
35. A closed cubical box is completely filled with water and is moving horizontally towards right with an acceleration a . The resultant normal force by the water on the top of the box.
 (1) Passes through the centre of the top
 (2) Passes through a point to the right of the centre
 (3) Passes through a point to the left of the centre
 (4) Becomes zero
36. Water is flowing through a long horizontal tube. Let P_A and P_B be the pressure at two point A and B of the tube, then -
 (1) P_A must be equal to P_B
 (2) P_A must be greater than P_B
 (3) P_A must be less than P_B
 (4) $P_A = P_B$ only if cross-sectional area at A and B are equal
37. An ice cube is suspended in vacuum in a gravity free hall. As ice melt, it
 (1) Will retain its cubical shape
 (2) Will change its shape to spherical
 (3) Will fall down on the floor of the hall
 (4) will fly up
38. When water droplets merge to form a bigger drop:
 (1) Energy is liberated
 (2) Energy is absorbed
 (3) Energy is neither liberated nor absorbed
 (4) Energy may either be liberated or absorbed depending on the nature of the liquid

39. Which of the following graphs may represent the relation between the capillary rise (h) and the radius (r) of the capillary?



- (1) a (2) b (3) c (4) d
40. Viscosity is a property of -
- (1) Liquids only
(2) Solids only
(3) Solids and liquids only
(4) Liquids and gases only
41. When a capillary tube is dipped into a liquid, the liquid neither rises nor falls in the capillary if :
- (a) The surface tension of the liquid must be zero
(b) The contact angle must be 90°
(c) The surface tension of the liquid may be zero
(d) The contact angle may be 90°
- (1) a, b (2) b, c (3) c, d (4) a, d
42. When contact angle in capillary tube is obtuse, then pressure just below the free surface of liquid in the capillary tube is :-
- (1) More than atmospheric pressure
(2) Less than atmospheric pressure
(3) Equal to atmospheric pressure
(4) None of these
43. A rigid ring A and a rigid thin disk B both made of same material, when gently placed on water, just manage to float due to surface tension as shown in the figure. Both the ring and the disk have same radius. What can you conclude about their masses (thickness may be different)?



- (1) Both have the same mass.
(2) Mass of the ring is half of that of the disk.
(3) Mass of the ring is double to that of the disk.
(4) More information is needed to decide.

44. The grease deposited on a glass plate can be easily removed by cleaning the glass with hot water containing detergent powder, because the detergent powder :-

- (1) Reduces the angle of contact between the solution and glass
(2) Increases the temperature of the solution
(3) Decrease the density of the solution
(4) Makes the angle of contact between solution and the glass to an obtuse angle

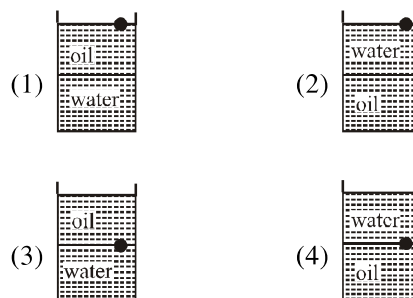
45. The graph between terminal velocity (along y-axis) and square of radius (along x-axis) of spherical body of density ρ allowed to fall through a fluid of density σ is a :-

- (1) Straight line with positive slope
(2) Straight line with negative slope
(3) Straight line perpendicular to x-axis
(4) Straight line perpendicular to y-axis

46. A fluid is undergoing steady flow. Therefore:

- (1) the velocity of any given molecule of fluid does not change
(2) the pressure does not vary from point to point
(3) the velocity at any given point does not vary with time
(4) the density does not vary from point to point

47. A ball is made of a material of density ρ where $\rho_{\text{water}} > \rho > \rho_{\text{oil}}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position ?



48. The relative density of a metal may be found by hanging a block of the metal from a spring balance and noting that in air the balance reads (5 ± 0.05) N while in water it reads (4 ± 0.05) N. The relative density would be quoted as:

- (1) (5 ± 0.05) (2) $5 \pm 11\%$
 (3) (5 ± 0.10) (4) $5 \pm 6\%$

49. You do a very precise experiment to test the properties of a piece of wire. You stretch the wire by applying an outward force to each end of the wire and measure its change in length. If you then precisely double this force and the wire still remains elastic but is beyond proportionality limit, the change in length of the wire

- (1) will exactly double
 (2) will just slightly more than double
 (3) will just slightly less than double
 (4) will decrease by exactly a factor of two.

50. Arrangement of three metallic rods riveted to each other is as shown in figure. α_{Al} is the coefficient of linear expansion for aluminum and α_s (when $\alpha_{Al} > \alpha_s$) for steel. The arrangement is heated, then

Steel
Al
Steel

- (1) Tensile stress is developed in the aluminum rod.
 (2) Compressive stress is developed in the aluminum rod
 (3) No stress is developed in the aluminum rod
 (4) Data's are incomplete to decide